

Collecting Data from the Web

The Statistical and Epistemological Foundations of Synthesis

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The Fixed-Effect Model: Assumptions and Derivation

The fixed-effect model assumes that all included studies estimate a single common true effect θ .

The pooled estimate is the inverse-variance weighted mean:

$$\hat{\theta} = \frac{\sum w_i \hat{\theta}_i}{\sum w_i}$$

where $w_i = 1/v_i$ and v_i is the within-study variance.

The Critical Assumption

The model assumes between-study variance is zero. Any observed difference between studies is attributed entirely to sampling error. In systematic reviews across diverse populations, this assumption is almost always violated.

The Random-Effects Model: DerSimonian–Laird

When studies differ in populations or interventions, the true effect is not a single value but a distribution.

The random-effects model (DerSimonian & Laird, 1986) adds a between-study variance component τ^2 to the weight:

$$w_i^* = \frac{1}{v_i + \hat{\tau}^2}$$

Implications:

- Small studies are penalised less than in the fixed-effect model.
- Confidence intervals for the pooled estimate are wider.
- This model is the default choice for most systematic reviews due to inherent clinical and methodological heterogeneity.

Effect Size Measures: A Taxonomy

Meta-analysis requires converting disparate study outcomes into a common metric.

Family	Measures
Continuous	Standardised Mean Difference (Cohen's d , Hedges' g)
Binary	Odds Ratio (OR), Risk Ratio (RR), Risk Difference (RD)
Associational	Correlation Coefficient (r , Fisher's z)
Observational	Adjusted Regression Coefficients

Note: Binary measures like OR and RR must be log-transformed before pooling to ensure symmetry around the null (1.0).

Variance Estimation and Small-Sample Corrections

A pooled estimate is only as good as the variance estimates of the primary studies.

- **Small-sample bias:** Cohen's d overestimates the true effect in small samples; Hedges' g applies a correction factor.
- **The Delta Method:** Used to approximate the variance of transformed effect sizes (e.g., log odds ratios).

The Imputation Problem

Abstracts often report effect sizes without standard errors or confidence intervals. Imputing these missing variances from p -values or sample sizes is mathematically possible but introduces substantial structural risk.

Measuring Heterogeneity: Q , I^2 , and τ^2

Three complementary statistics quantify between-study heterogeneity:

- 1 **Cochran's Q :** A χ^2 statistic testing the null hypothesis of homogeneity. Highly underpowered when the number of studies is small.
- 2 **I^2 Statistic (Higgins & Thompson, 2002):** The proportion of total variance attributable to between-study variance, rather than sampling error.
- 3 **τ^2 :** The absolute measure of between-study variance in the metric of the effect size.

Rule of thumb: $I^2 > 50\%$ indicates substantial heterogeneity, but I^2 can be artificially inflated in meta-analyses of very large observational studies.

Interpreting Heterogeneity: Clinical vs Statistical

Statistical heterogeneity ($I^2 > 50\%$) does not automatically invalidate a meta-analysis—it demands investigation.

The Critical Distinction:

- **Statistical heterogeneity:** Variability in effect sizes beyond what is expected by chance.
- **Clinical heterogeneity:** Variability in populations, interventions, and outcomes.

Epistemological Note

A high I^2 in a well-conducted meta-analysis is often a substantively interesting finding, not a methodological failure. It signals that the intervention's effect depends heavily on context.

Subgroup Analysis and Meta-Regression

When heterogeneity is high, the analyst must investigate its sources.

Subgroup Analysis

Stratifying the meta-analysis by a categorical moderator (e.g., study design, country income level). Tests whether the pooled effect differs between groups.

Meta-Regression

Modelling the effect size as a function of continuous moderators (e.g., mean age, year of publication). Tests whether the effect size changes systematically.

The Risk: Data-dredging. Post-hoc subgroup analyses are highly prone to false positives. Moderators must be pre-registered in the review protocol.

Risk of Bias in Randomised Trials: The RoB 2 Tool

The Cochrane Risk of Bias 2 (RoB 2) tool (Sterne et al., 2019) is the standard for assessing validity in RCTs.

Five Domains of Bias:

- 1 Bias arising from the randomisation process.
- 2 Bias due to deviations from intended interventions.
- 3 Bias due to missing outcome data.
- 4 Bias in measurement of the outcome.
- 5 Bias in selection of the reported result.

Each domain is rated *Low*, *Some Concerns*, or *High*. The overall risk of bias is determined by the highest risk across any domain.

Risk of Bias in Observational Studies: ROBINS-I

Observational studies require a fundamentally different framework from RCTs.

The **ROBINS-I** tool (Sterne et al., 2016) assesses bias by comparing the observational study to a hypothetical "target trial".

Key Domains:

- **Pre-intervention:** Confounding, selection bias.
- **At intervention:** Bias in classification of interventions.
- **Post-intervention:** Deviations from intended interventions, missing data, measurement bias, reporting bias.

Incorporating Risk of Bias into the Synthesis

How should risk of bias assessments affect the meta-analysis?

Three Approaches:

- ① **Exclusion:** Restrict the primary analysis strictly to low-risk studies. (Often leaves too few studies).
- ② **Sensitivity Analysis:** Compare the pooled estimate with and without high-risk studies. (The standard approach).
- ③ **Weighting:** Down-weight high-risk studies in the pooled estimate. (Methodologically controversial).

Ultimately, the overall risk of bias feeds into the GRADE assessment, determining the certainty of the final evidence.

Detecting Publication Bias: Funnel Plots and Egger's Test

The Funnel Plot: A scatter plot of effect size against standard error (or sample size). Under no publication bias, the plot should be symmetric around the pooled estimate.

- **Asymmetry** suggests that small studies with null or negative results are missing (the file-drawer problem).
- **Egger's Test** (Egger et al., 1997): A formal regression test of funnel plot asymmetry.

Limitation

Asymmetry can also be caused by genuine heterogeneity (e.g., small studies use different populations or higher doses), not just publication bias.

Correcting for Publication Bias

If publication bias is detected, how do we estimate the true effect?

- **Trim-and-Fill** (Duval & Tweedie, 2000): Iteratively removes the most extreme small studies causing asymmetry, re-estimates the centre, and then "fills in" missing mirror-image studies.
- **Selection Models** (Vevea & Hedges, 1995): Model the probability that a study is published as a function of its p -value.

These corrections are mathematically elegant but rely on strong assumptions. They should be reported as sensitivity analyses, never as the primary pooled estimate.

Advanced Methods: Network Meta-Analysis (NMA)

Standard pairwise meta-analysis can only compare two interventions at a time.

Network Meta-Analysis (Lumley, 2002) synthesises evidence from a network of direct and indirect comparisons, allowing simultaneous ranking of multiple interventions.

The Transitivity Assumption

NMA assumes that indirect comparisons (A vs C via B) are valid. This requires that the studies comparing A vs B and B vs C are sufficiently similar in populations and settings.

Advanced Methods: Bayesian Meta-Analysis

The Bayesian alternative to the frequentist DerSimonian–Laird estimator incorporates prior knowledge and produces more interpretable probability statements.

- **Priors on τ^2 :** Using half-normal or half-Cauchy priors stabilises the between-study variance estimate when the number of studies is small.
- **Posterior Predictive Distribution:** The expected effect in a new, as-yet-unobserved study.

Bayesian credible intervals have a natural interpretation: "There is a 95% probability that the true pooled effect lies within this interval"—a statement frequentist confidence intervals cannot legally make.

Synthesis Without Meta-Analysis (SWiM)

When quantitative pooling is inappropriate, the SWiM framework (Campbell et al., 2020) provides a structured approach to narrative synthesis.

The Four SWiM Components:

- 1 Grouping and coding the studies.
- 2 Transforming data within studies.
- 3 Translating data within studies.
- 4 Synthesising data across studies.

Vote-counting by direction of effect is a last resort. "Narrative synthesis" is not simply a description of individual studies; it is a structured, reproducible analytical procedure.

The GRADE Framework: From Evidence to Recommendations

GRADE (Guyatt et al., 2008) is the international standard for rating the certainty of a body of evidence.

Factors that lower certainty:

- Risk of bias
- Inconsistency (heterogeneity)
- Indirectness (PICO mismatch)
- Imprecision (wide CIs)
- Publication bias

Factors that raise certainty:

- Large magnitude of effect
- Dose-response gradient
- Plausible confounding would have reduced the effect

Certainty Levels: High, Moderate, Low, Very Low.

Sensitivity Analysis: The Robustness of the Estimate

A pooled estimate is only credible if it is robust to reasonable changes in analytical decisions.

Standard Sensitivity Analyses:

- Choice of statistical model (fixed vs random effects).
- Inclusion vs exclusion of high-risk-of-bias studies.
- Imputation of missing variances.
- **Leave-one-out analysis:** Re-computing the pooled estimate after removing each study in turn to identify highly influential outliers.

The Epistemological Status of a Pooled Estimate

A critical reflection on what a meta-analytic pooled estimate actually represents.

The Ecological Fallacy in Meta-Analysis

The pooled estimate is not the "true" effect for any specific patient or setting. It is the mean of a distribution of effects across a hypothetical population of studies.

- **Statistical vs Clinical Significance:** A statistically significant pooled estimate with a narrow confidence interval can still be practically irrelevant if the absolute magnitude is trivial.

Pre-registration and the Evidentiary Record

The credibility of the synthesis depends on decisions made before data collection.

Pre-registration (PROSPERO / OSF): The research question, eligibility criteria, extraction schema, statistical analysis plan, and planned subgroup analyses must be publicly registered. Post-hoc deviations must be explicitly disclosed.

The Final Evidentiary Record:

- 1 The data extraction table.
- 2 The risk of bias assessments.
- 3 The forest plot(s) with heterogeneity statistics.
- 4 The PRISMA 2020 flow diagram.
- 5 The GRADE summary of findings table.